

Writing Your Own Security Tools

Why Build Your Own

Readiness + understanding. You'll use Nmap, hashcat, and Burp daily – but knowing how a scanner works demystifies evasion and detection.

A Simple Port Scanner

```
import socket

def scan(host, ports):
    for port in ports:
        s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
        s.settimeout(1)
        try:
            s.connect((host, port))
            print(f"{host}:{port} OPEN")
        except (socket.timeout, ConnectionRefusedError):
            pass # filtered or closed
        finally:
            s.close()

scan("10.0.0.5", [22, 80, 443, 3389, 445])
```

Banner Grabbing (Recon Gold)

```
import socket

def grab_banner(host, port):
    s = socket.socket()
    s.settimeout(5)
    try:
        s.connect((host, port))
        s.send(b"HEAD / HTTP/1.1\r\nHost: " + host.encode() + b"\r\n\r\n")
        print(f"{host}:{port} ->", s.recv(1024).decode(errors="ignore").strip())
    except Exception as e:
        print(f"{host}:{port} -> {e}")
```

```
finally:  
    s.close()
```

Banners reveal the exact software + version → exploit search (searchsploit) for the version.

A Mini Hash Cracker (Wordlist)

```
import hashlib  
  
target = "5f4dcc3b5aa765d61d8327deb882cf99" # md5("password")  
with open("rockyou-small.txt", "r", encoding="latin-1") as f:  
    for word in f:  
        w = word.strip()  
        if hashlib.md5(w.encode()).hexdigest() == target:  
            print("CRACKED:", w)  
            break  
    else:  
        print("Not in wordlist")
```

Why professionals use hashcat instead: GPU acceleration + millions of guesses/sec vs a Python loop at thousands/sec. But seeing the loop teaches the concept.

A Log Analyzer (Blue Team)

```
import re, collections  
  
pattern = re.compile(r'(\d+\.\d+\.\d+\.\d+)')  
failures = collections.Counter()  
  
for line in open("access.log"):  
    if "401" in line: # unauthorized  
        m = pattern.search(line)  
        if m: failures[m.group(1)] += 1  
  
for ip, count in failures.most_common(10):  
    print(count, ip)
```

Ethics Warning (Always)

Building these tools is legal and good. Using them against systems you don't own and don't have written permission to test is illegal. Lab-only. Always.

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